

Referee Report

*Significance Bias in Government Intervention Research: Evidence from
Top Finance Journals*

Submitted to the *Journal of Finance*

Referee #2

Anonymous

May 2026

Summary Recommendation

Decision: Reject — Encourage Resubmission

The paper addresses an important and timely question: does the published finance literature on government intervention systematically underrepresent null findings? If true, this would have significant implications for regulatory policy, which relies heavily on the academic evidence base. The topic is well-motivated, the writing is generally clear, and the empirical exercise is transparent. However, the paper has several fundamental identification problems that prevent me from recommending acceptance at this stage. The core issue is that the research design compares two populations that are never directly observed at the same stage of the research pipeline, which introduces confounds that the authors have not adequately addressed. I describe these concerns below in order of severity.

1 Major Concerns

1. The Fundamental Identification Problem: Two Non-Overlapping Samples

The paper’s identification strategy rests on comparing publication rates across direction codes within a combined sample of NBER working papers and published journal articles. The critical design choice is that only 8 of 183 published papers (4.4%) are matched to an NBER working paper. This means the sample consists almost entirely of two non-overlapping populations:

- 175 published journal articles with *no* observed pre-publication counterpart.
- 113 NBER working papers with *no* confirmed journal publication.

The paper’s identifying assumption is that the NBER sample represents the pre-publication distribution of findings and that the journal sample represents the post-publication distribution. But because these are *different papers* rather than the same papers tracked before and after the editorial filter, the comparison is confounded. The 113 unpublished NBER papers and the 175 published journal papers may differ along many dimensions other than direction code—including research quality, author seniority, institutional affiliation, identification strategy, and data quality. If null-finding papers are of lower average quality (e.g., because genuinely null results in this domain are harder to publish credibly due to underpowering or weaker identification), the 36% publication rate for null papers reflects a quality gap, not a significance filter. The paper does not control for any of these alternative explanations.

The gold-standard design for this question would track the same papers from pre-submission through editorial decision—as in *Franco2014*’s survey of NSF-funded studies, where grant records identify research regardless of whether it was ultimately published. The NBER working paper approach approximates this, but the near-zero match rate means the approximation is very rough. The authors should either (a) dramatically improve the matching rate, (b) add controls for observable quality proxies, or (c) substantially qualify the causal interpretation.

2. NBER Selection Is Not a Random Draw from Research Output

The paper acknowledges that “NBER membership is itself selective” but argues the selection “operates largely on researcher quality and institution rather than on the direction of findings.” This may be true in the cross-section, but it creates a severe problem for the within-researcher inference the paper requires.

Specifically: NBER-affiliated researchers are among the most productive and well-connected academics in finance and economics. Their null-finding papers (which constitute 75.2% of the NBER sub-sample) may be substantially more publishable than null-finding papers from the broader population. If this is so, the 36% publication rate for null-finding NBER papers represents an *upper bound* on the true publication probability for null results in the field. The paper’s policy conclusion that “the published record overstates the share of interventions with detectable effects” would then be conservative—but the magnitude of the bias would also be smaller than reported. More problematically, there is no way to assess from the current design whether the null papers in the NBER sample represent genuinely null interventions or simply research programs that were ultimately abandoned because early results were weak.

3. LLM Direction Coding: Validity and Potential Systematic Error

The direction codes that drive all results are assigned by an LLM reading the abstract of each paper. Three concerns arise:

(a) **No systematic human validation.** The paper mentions “a held-out set” in the Related Literature section but provides no validation statistics—no confusion matrix, no inter-rater agreement, no false-positive or false-negative rates for direction coding. Without these, readers cannot assess whether the LLM’s direction assignments are reliable. A 10% error rate applied asymmetrically—for example, if the LLM more often miscodes ambiguous positives as null than ambiguous negatives—could substantially affect the estimated premiums.

(b) **Abstract language is itself affected by the publication process.** Pub-

lished abstracts are polished through the editorial process; NBER working paper abstracts are often rougher and less definitive in their framing. If published papers are written to emphasize the significance and direction of their results (which they are), the LLM may systematically assign stronger direction codes to published papers and weaker or null codes to NBER papers. This would mechanically generate the observed publication-rate gap even in the absence of any true significance filter.

(c) **The definition of “positive” is normatively charged.** A paper finding that capital requirements reduce bank risk-taking is coded positive (+1) because “the intervention achieved its intended regulatory goal.” But many economists would code the same result as “the intervention constrained banks” and ask whether the regulatory goal was desirable. This coding convention is not neutral, and it may map imperfectly onto what editors and referees consider a “positive” finding. The authors should test the sensitivity of their results to alternative coding conventions.

4. Quality Differences Between Null and Directional Papers

The most important alternative explanation for the null-paper disadvantage is that null-finding papers in this domain are, on average, of lower empirical quality. Credibly estimating a null effect of a government policy requires the same statistical power and identification rigor as estimating a positive or negative effect—and in a domain where many interventions have genuinely large effects, a finding of no effect is inherently harder to make credibly. Editors and referees may rationally discount null results more heavily because they are more likely to reflect underpowering than to reflect a true zero.

The paper does not control for any proxy of research quality: not identification strategy (DiD vs. OLS), not number of citations, not author quality, not data quality, not sample size, not the journal tier in which NBER authors typically publish. Until these confounds are addressed, the significance-bias interpretation is not the only—or even the most plausible—explanation for the observed pattern.

5. Inconsistency in Reporting the JF Results

The paper makes a central claim that “the finance publication process exhibits strong significance bias.” However, the journal-level results tell a more nuanced story that is not adequately reconciled. The JF coefficients are labeled 1.022*** and 1.067*** in Table 4, but the text then states they “fall short of statistical significance at conventional levels” and attributes this to low power. There is a contradiction: triple-star notation conventionally indicates $p < 0.01$, yet the text implies the JF estimates are not statistically significant. The authors should carefully verify and reconcile the significance labels in Table 4.

More substantively, the fact that the largest journal in the field—and the journal to which this paper is submitted—shows no significant significance bias in this domain is a material finding that deserves more prominent discussion rather than being explained away with a power argument.

2 Moderate Concerns

6. Sample Size and Statistical Power

The analysis rests on 296 papers, of which 113 are NBER-only and 183 are published. The journal-level regressions have 151 (JF), 180 (RFS), and 191 (JFE) observations, but within each journal the published sub-sample is only 38, 67, and 78 papers, respectively. The sub-period regressions have 97 and 154 observations. These are small samples for probit models with multiple parameters. The authors should report that all results are verified with linear probability models (which avoid separation issues in small samples) and consider reporting bootstrap confidence intervals.

7. Interpretation of the 2000–2007 Sub-Period Omission

The authors omit the 2000–2007 sub-period because it has only 45 papers “insufficient for reliable probit estimation.” But 45 papers is not an unreasonably small sample for a bivariate probit, and the omission of this sub-period—which covers the pre-crisis period when government intervention was arguably less salient—could affect the time-

trend interpretation. At minimum, the 2000–2007 results should be reported (with appropriate caveats) rather than simply dropped.

8. The Corpus Description Is Internally Inconsistent

The introduction states the authors screen “a corpus of more than 40,000 documents,” but the Data section reports a total corpus of 33,224 documents. This discrepancy should be resolved. (I note that the inconsistency appears to stem from earlier draft language referring to the 1990–2024 period; if the paper has been revised to cover only 2000–2024, all corpus references must be updated consistently.)

9. Robustness Check R4 Is Misspecified

Robustness check R4 restricts to the 169 high-confidence LLM classifications and reports that the subsample is endogenously selected (88% published, versus 62% in the full sample). The authors correctly note that this renders the coefficients uninformative and say they “do not draw conclusions” from the exercise. If that is the case, the check adds no information and should either be replaced with a meaningful sensitivity analysis or omitted.

3 Minor Comments

1. **The NBER-to-journal match rate of 4.4% is surprisingly low.** The paper attributes this to title revisions and non-NBER publication paths. The authors should quantify how many published papers they believe had NBER working paper precursors that were simply not matched (perhaps by checking known papers by hand), and assess whether the 175 unmatched published papers are a representative cross-section of the direction distribution.
2. **No comparison to other domains.** The paper would be strengthened considerably by a comparison sample—either from a different domain of finance (e.g., corporate governance without explicit policy intervention) or from another field

(e.g., labor economics, where similar studies exist). Without a comparison, it is hard to assess whether the documented $2.3\times$ ratio is large or typical.

3. **The discussion of mechanisms is speculative.** The paper interprets the 2008–2015 negative-finding premium as “elevated editorial demand for research documenting the unintended costs of the post-crisis regulatory expansion.” This is a plausible narrative but is not tested. Editor letters, referee reports, or desk-rejection data could provide evidence for or against this mechanism.
4. **No correction for multiple testing.** The paper runs multiple probit specifications, journal subsets, and sub-period regressions. Given the exploratory nature of some of these tests, Bonferroni or Benjamini–Hochberg corrections (or at minimum a discussion of multiple testing concerns) would strengthen the paper.
5. **Terminology.** The term “significance bias” is used throughout, but this term has a specific meaning in the statistics literature (bias in test statistics near the significance threshold, as in *Brodeur2016*). The phenomenon the authors document is more precisely described as a “publication filter” or “null-result penalty.” Using established terminology or carefully defining the authors’ usage would reduce confusion.
6. **References.** The paper cites *Ash2023* for the use of LLMs in economics. Given the rapid growth of this literature, the authors should also cite more recent work on LLM reliability and potential biases in text classification tasks, particularly for economics applications.

Summary of Required Revisions

For this paper to be publishable in a top journal, the authors must:

1. Address the fundamental identification problem by substantially improving the NBER-to-journal match rate, adding controls for observable quality, or obtaining an alternative data source (e.g., editor records, submission-level data from a journal) that allows tracking the same papers through the publication process.

2. Provide systematic LLM validation statistics (confusion matrix, inter-rater reliability with human coders) for both the in-scope classification and the direction coding.
3. Reconcile the JF significance labels in Table 4 with the text discussion.
4. Report results for the 2000–2007 sub-period.
5. Add controls for paper quality (author citations, identification strategy, sample size) or explicitly test whether the null-paper disadvantage survives quality controls.
6. Replace or remove Robustness Check R4, which by the authors’ own admission yields uninformative estimates.

The topic is important and the paper is well-executed at the data-collection and descriptive level. The authors have built an impressive pipeline. I encourage a revision that addresses the identification concerns, particularly items 1–3 above.

Note: References in this report follow the style of the manuscript under review. “Brodeur2016” refers to Brodeur, Lé, Sangnier, and Zylberberg (2016); “Franco2014” refers to Franco, Malhotra, and Simonovits (2014); “Andrews2019” refers to Andrews and Kasy (2019); “Ash2023” refers to Ash et al. (2023).